

Product and Quotient rule

In this session³⁰ we are going to learn about the product and quotient rule for derivatives. This rule allow us to compute the derivative of a product of functions. Assuming that we have a function $f(x) = u(x)v(x)$, the derivative of the function can be computed as follows,

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$$\frac{df(x)}{dx} = \frac{d}{dx} [u(x)v(x)] = \frac{du(x)}{dx} v(x) + u(x) \frac{dv(x)}{dx}. \quad (3)$$

For example, if we need to compute the derivative of $f(x) = x^2 \sin(x)$, we can compute it as follows,

$$\begin{aligned} \frac{d}{dx} [x^2 \sin(x)] &= \left[\frac{d}{dx} x^2 \right] \sin(x) + x^2 \left[\frac{d}{dx} \sin(x) \right] \\ &= [2x] \sin(x) + x^2 [\cos(x)] \\ &= 2x \sin(x) + x^2 \cos(x). \end{aligned}$$

On the other hand, the quotient rule allow us to easily compute the derivative of a quotient of functions. In this instance the function is of the form $f(x) = \frac{u(x)}{v(x)}$. In order to compute it's derivative we use the following rule,

$$\frac{df(x)}{dx} = \frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{1}{v^2(x)} \left[\frac{du(x)}{dx} v(x) - u(x) \frac{dv(x)}{dx} \right]. \quad (4)$$

For example, if we are asked to calculate the derivative of $f(x) = \frac{x^2}{\sin(x)}$ we can do it as follows,

$$\begin{aligned} \frac{df(x)}{dx} &= \frac{1}{\sin^2(x)} \left[\left(\frac{d}{dx} x^2 \right) \sin(x) - x^2 \left(\frac{d}{dx} \sin(x) \right) \right] \\ &= \frac{1}{\sin^2(x)} [(2x) \sin(x) - x^2 (\cos(x))] \\ &= \frac{2x}{\sin(x)} - \frac{x^2}{\sin(x)} \cot(x) = x \csc(x) [2 - x \cot(x)] \end{aligned}$$

Class activity

Compute the derivatives using the product, quotient and power rules of the following functions. You can also use the rules for trascendental functions if necessary.

$$\begin{aligned} f(x) &= x^3 e^x \\ g(x) &= \frac{\sin(x)}{x} \\ h(x) &= x^2 \ln(3x) \end{aligned}$$